

# $n$ -SCRAMBLED CANTOR SETS

B. MILLER

ABSTRACT. For all  $n \geq 2$ , we produce a homeomorphism of Cantor space that does not have an  $(n + 1)$ -scrambled Cantor set but does have a single uncountable set which is  $m$ -scrambled for all  $m \geq n + 1$  and whose closure is an  $n$ -scrambled Cantor set.

## INTRODUCTION

Given a metric space  $X$ ,  $f: X \rightarrow X$ , and  $n \geq 2$ , a sequence  $x \in X^n$  is *proximal* if  $\liminf_{i \rightarrow \infty} \max_{j < k < n} d_X(f^i(x(j)), f^i(x(k))) = 0$ , *separated* if  $\limsup_{i \rightarrow \infty} \min_{j < k < n} d_X(f^i(x(j)), f^i(x(k))) > 0$ , and *Li–Yorke* if it is proximal and separated. A set  $Y \subseteq X$  is  *$n$ -scrambled* if every injective sequence in  $Y^n$  is Li–Yorke. We say that  $Y$  is  *$(<\mathbb{N})$ -scrambled* if it is  $n$ -scrambled for all  $n \geq 2$ .

In [GGM21], it was shown that uncountable 2-scrambled sets yield 2-scrambled Cantor sets in Polish dynamical systems. Here we show:

**Theorem 1** (ZFC). *Suppose that  $n \geq 2$ . Then there is a homeomorphism of  $2^{\mathbb{N}}$  that does not have an  $(n + 1)$ -scrambled perfect set but does have an uncountable  $(<\mathbb{N})$ -scrambled set whose closure is an  $n$ -scrambled Cantor set.*

Let  $\text{INJ}_X^n$  denote the set of injective sequences of elements of  $X$  of length  $n$ . Let  $\text{LY}_f$  denote the set of finite Li–Yorke sequences. A *graph* on  $X$  is an irreflexive symmetric set  $G \subseteq X \times X$ . We say that  $Y$  is a  *$G$ -clique* if its distinct points are  $G$ -related, and  *$G$ -independent* if its points are not  $G$ -related. Let  $\text{CLQ}_G^{\geq n}$  denote the set of finite injective sequences of length at least  $n$  whose images are  $G$ -cliques, and let  $\text{IND}_G^n$  denote the set of injective sequences of length  $n$  whose images are  $G$ -independent. A *homomorphism* from a relation  $R$  on  $X$  to a relation  $S$  on  $Y$  is a function  $\pi: X \rightarrow Y$  whose compositions with sequences in  $R$  are in  $S$ . Such a function is a *homomorphism* from  $(R, R')$  to  $(S, S')$  if it is also a homomorphism from  $R'$  to  $S'$ . The *saturation* of a set  $Y \subseteq X$  under a bijection  $T: X \rightarrow X$  is  $[Y]_T = \bigcup_{i \in \mathbb{Z}} T^i(Y)$ . A subset

---

2020 *Mathematics Subject Classification*. Primary 37B05; Secondary 03E15.

*Key words and phrases*. Li–Yorke chaos, scrambled set, topological dynamics.

of a topological space is  $F_\sigma$  if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is:

**Theorem 2** (ZF + DC). *Suppose that  $G$  is an  $F_\sigma$  graph on  $2^\mathbb{N}$  and  $n \geq 2$ . Then there exist a homeomorphism  $T: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$  and a continuous injective homomorphism  $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$  from  $(\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^\mathbb{N}}^n, \text{IND}_G^{n+1})$  to  $(\text{LY}_T, \sim \text{LY}_T)$  for which  $|\sim[\pi(2^\mathbb{N})]_T| = 1$ .*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

### 1. EMBEDDING HYPERGRAPHS

Define  $S: 2^\mathbb{Z} \rightarrow 2^\mathbb{Z}$  by  $S(c)(i) = c(i+1)$  for all  $i \in \mathbb{Z}$  and  $c \in 2^\mathbb{Z}$ . Let  $\text{SEP}_f$  denote the set of finite separated sequences. Theorem 2 follows from the usual topological characterization of  $2^\mathbb{N}$  and:

**Theorem 1.1** (ZF + DC). *Suppose that  $G$  is an  $F_\sigma$  graph on  $2^\mathbb{N}$  and  $n \geq 2$ . Then there exist an  $S$ -invariant perfect set  $F \subseteq 2^\mathbb{Z}$ , whose finite sequences are proximal, and a continuous injective homomorphism  $\pi: 2^\mathbb{N} \rightarrow F$  from  $(\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^\mathbb{N}}^n, \text{IND}_G^{n+1})$  to  $(\text{SEP}_S, \sim \text{SEP}_S)$  for which  $0^\mathbb{Z}$  is the unique element of  $F \setminus [\pi(2^\mathbb{N})]_S$ .*

*Proof.* For all  $s \in 2^{<\mathbb{N}}$ , set  $\mathcal{N}_s = \{c \in 2^\mathbb{N} \mid \forall i < |s| \ s(i) = c(i)\}$ . For all  $S \subseteq 2^{<\mathbb{N}}$ , set  $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$ . For all  $\ell \in \mathbb{N}$ ,  $n' \geq n$ , and  $p: 2^\ell \rightarrow 2^{n'}$ , set  $\mathcal{N}_p = \prod_{i < n'} \mathcal{N}_{p^{-1}(\{i\})}$ . We say that  $p$  is a *near bijection* if it induces a bijection of  $2^\ell \setminus p^{-1}(0)$  with  $n' \setminus \{0\}$  but  $\{0^\ell\} \subsetneq p^{-1}(\{0\})$ . For all  $0 < k < n'$ , let  $p^{-1}(k)$  denote the unique element of  $p^{-1}(\{k\})$ .

Fix an increasing sequence of closed graphs  $G_m$  on  $2^\mathbb{N}$  whose union is  $G$ . Define  $G_{\ell,m} = \{(c, c') \in 2^\mathbb{N} \times 2^\mathbb{N} \mid G_m \cap (\mathcal{N}_{c \upharpoonright \ell} \times \mathcal{N}_{c' \upharpoonright \ell}) \neq \emptyset\}$ . For all  $\ell \in \mathbb{N}$ , let  $P_\ell$  be the set of all near bijections  $p: 2^\ell \rightarrow n$ . For all  $\ell, m \in \mathbb{N}$  and  $n' > n$ , let  $P_{\ell,m,n'}$  be the set of near bijections  $p: 2^\ell \rightarrow n'$  such that  $\mathcal{N}_{p^{-1}(j)} \times \mathcal{N}_{p^{-1}(k)} \subseteq G_{\ell,m}$  for all  $0 < j < k < n'$ . Fix  $I \subseteq \mathbb{N} \setminus \{0\}$  and a bijection  $\phi: I \rightarrow \mathbb{N} \cup \bigcup_{\ell \in \mathbb{N}} \{\ell\} \times P_\ell \cup \bigcup_{\ell, m \in \mathbb{N}, n' > n} \{\ell\} \times \{m\} \times \{n'\} \times P_{\ell,m,n'}$  such that  $I \cap [i, 2i + w(i)] = \{i\}$  for all  $i \in I$ , where  $w(i) = 0$  if  $\phi(i) \in \mathbb{N}$ ,  $w(i) = n - 2$  if  $\phi(i) \in \{\ell\} \times P_\ell$  for some  $\ell \in \mathbb{N}$ , and  $w(i) = (m+1)(n'-2)$  if  $\phi(i) \in \{\ell\} \times \{m\} \times \{n'\} \times P_{\ell,m,n'}$  for some  $\ell, m \in \mathbb{N}$  and  $n' > n$ . Define a continuous function  $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{Z}$  by  $\pi(c)(i) = 1$  if and only if

- (1)  $i = 0$ ;
- (2)  $i \in I$ ,  $\phi(i) \in \mathbb{N}$ , and  $c(\phi(i)) = 1$ ;
- (3)  $\exists 0 < k < n \exists \ell \in \mathbb{N} \exists p \in P_\ell$   
 $((\ell, p) = \phi(i - (k - 1)) \text{ and } p(c \upharpoonright \ell) = k)$ ; or
- (4)  $\exists n' > n \exists 0 < k < n' \exists \ell, m \in \mathbb{N} \exists p \in P_{\ell,m,n'}$   
 $((\ell, m, n', p) = \phi(i - (m + 1)(k - 1)) \text{ and } p(c \upharpoonright \ell) = k)$ .

**Lemma 1.2.**  $\pi$  is injective.

*Proof.* Suppose that  $c, c' \in 2^{\mathbb{N}}$  are distinct. Then there exists  $k \in \mathbb{N}$  with  $c(k) \neq c'(k)$ . Fix  $i \in I$  for which  $\phi(i) = k$ . As  $\pi(c)(i) = c(k)$  and  $\pi(c')(i) = c'(k)$ , it follows that  $\pi(c)(i) \neq \pi(c')(i)$ , so  $\pi(c) \neq \pi(c')$ .  $\square$

The *product metric* on  $2^{\mathbb{Z}}$  is given by  $\rho(c, c') = \sum_{i \in \mathbb{Z}} |c(i) - c'(i)|/2^{|i|}$  for all  $c, c' \in 2^{\mathbb{Z}}$ . Define  $N_i = [i, i + w(i)]$  for all  $i \in I$ .

**Lemma 1.3.**  $\pi$  is a homomorphism from  $\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^{\mathbb{N}}}^n$  to  $\text{SEP}_S$ .

*Proof.* Suppose that  $c \in (2^{\mathbb{N}})^{n'}$  is in  $\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^{\mathbb{N}}}^n$ . By permuting the coordinates of  $c$ , we can assume that  $c(0) = 0^{\mathbb{N}}$  if  $0^{\mathbb{N}} \in c(n')$ . Suppose that  $\ell \in \mathbb{N}$  is sufficiently large that  $\text{proj}_{2^\ell} \circ c$  is injective,  $c(k) \upharpoonright \ell \neq 0^\ell$  for all  $0 < k < n'$ , and  $n' < 2^\ell$ .

If  $n' = n$ , then let  $p$  be the near bijection extending  $(\text{proj}_{2^\ell} \circ c)^{-1}$  in  $P_\ell$ . Fix  $i \in I$  such that  $\phi(i) = (\ell, p)$ . Then  $\text{proj}_{2^{N_i}} \circ \pi \circ c$  is injective, so  $\min_{j < k < n} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{n-2}$ .

If  $n' > n$ , then fix  $m \in \mathbb{N}$  such that  $c(j) G_m c(k)$  for all  $j < k < n'$ , and let  $p$  be the near bijection extending  $(\text{proj}_{2^\ell} \circ c)^{-1}$  in  $P_{\ell, m, n'}$ . Fix  $i \in I$  such that  $\phi(i) = (\ell, m, n', p)$ . Then  $\text{proj}_{2^{N_i}} \circ \pi \circ c$  is injective, so  $\min_{j < k < n'} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(m+1)(n'-2)}$ .  $\square$

**Lemma 1.4.**  $\pi$  is a homomorphism from  $\text{IND}_G^{n+1}$  to  $\sim\text{SEP}_S$ .

*Proof.* Suppose that  $c \in (2^{\mathbb{N}})^{n+1}$  and  $\pi \circ c$  is separated. Fix  $\epsilon > 0$  such that  $I' = \{i' \in \mathbb{N} \mid \min_{j < k < n+1} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$  is infinite, as well as  $r \in \mathbb{N}$  such that  $\sum_{|i| > r} 1/2^{|i|} < \epsilon$ . By throwing out finitely many elements of  $I'$ , we can assume that for all  $i' \in I'$ , there is at most one  $i \in I$  such that  $[i' - r, i' + r] \cap N_i \neq \emptyset$ . Our choice of  $r$  then yields a unique such  $i \in I$ , in which case  $\text{proj}_{2^{N_i}} \circ \pi \circ c$  is injective. It follows that  $\phi(i)$  is of the form  $(\ell, m, n', p)$ , where  $(m+1)(n-1) \leq 2r$ , and there exist  $j < k < n+1$  such that  $c(j) G_{\ell, m} c(k)$ .

By passing to an infinite subset of  $I'$ , we can assume that the values of  $j$ ,  $k$ , and  $m$  are independent of  $i' \in I'$ . Then there are unboundedly many values of  $\ell$  corresponding to the remaining  $i' \in I'$ , since otherwise there are only finitely many possibilities for both  $n'$  and  $p$ , and therefore only finitely many possibilities for  $i$ , despite the fact that the distance between any two elements of  $I'$  corresponding to the same value of  $i$  is at most  $2r$ . Hence  $c(j) G_m c(k)$ , so  $c(j) G c(k)$ , thus  $c \notin \text{IND}_G^{n+1}$ .  $\square$

The closure  $F$  of the  $S$ -saturation of  $\pi(2^{\mathbb{N}})$  is clearly perfect.

**Lemma 1.5.** The sequence  $0^{\mathbb{Z}}$  is the unique element of  $F \setminus [\pi(2^{\mathbb{N}})]_S$ .

*Proof.* If  $(c_k)_{k \in \mathbb{N}}$  is a sequence of elements of  $2^{\mathbb{N}}$  and  $(i_k)_{k \in \mathbb{N}}$  is a strictly increasing sequence of elements of  $\mathbb{N}$ , then  $S^{-i_k}(\pi(c_k)) \rightarrow 0^{\mathbb{Z}}$ . And if  $S^{i_k}(\pi(c_k))$  converges to a sequence  $c \in 2^{\mathbb{Z}} \setminus \{0^{\mathbb{Z}}\}$ , then  $c(i) = 1$  for a unique  $i \in \mathbb{Z}$ , so it is in the  $S$ -orbit of  $\pi(0^{\mathbb{N}})$ .  $\square$

Lemma 1.5 implies that the union of the supports of finitely many elements of  $F$  is contained in a finite union of translates of  $\{0\} \cup \bigcup_{i \in I} N_i$ . But such sets necessarily have arbitrarily long gaps, so every finite sequence of elements of  $F$  is proximal.  $\square$

**Proposition 1.6** (ZF + DC). *Suppose that  $n \geq 3$ ,  $X$  and  $Y$  are Polish spaces,  $G$  is a Borel graph on  $X$ ,  $T: Y \rightarrow Y$  is a Borel automorphism,  $B \subseteq Y$  is an uncountable Borel  $n$ -scrambled set,  $\pi: X \rightarrow Y$  is an injective Borel homomorphism from  $\text{IND}_G^n$  to  $\sim \text{LY}_T$ , and  $[\pi(X)]_T$  is co-countable. Then  $G$  has a perfect clique.*

*Proof.* Fix  $i \in \mathbb{Z}$  for which  $B \cap (T^i \circ \pi)(X)$  is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set  $P \subseteq (T^i \circ \pi)^{-1}(B)$  that is either a  $G$ -clique or  $G$ -independent. But  $G$ -independence would contradict the fact that  $B$  is  $n$ -scrambled.  $\square$

## 2. $n$ -SCRAMBLED SETS

We begin this section with the proof of our primary result:

*Proof (of Theorem 1).* By [KV12, Theorem 2.1] and a simple Baire category argument, there is an  $F_\sigma$  graph  $G$  on  $2^{\mathbb{N}}$  with an uncountable clique  $Y \subseteq 2^{\mathbb{N}}$  but no perfect clique. By Theorem 2, there exist a homeomorphism  $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  and a continuous injective homomorphism  $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  from  $(\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^{\mathbb{N}}}^n, \text{IND}_G^{n+1})$  to  $(\text{LY}_T, \sim \text{LY}_T)$  for which  $|\sim[\pi(2^{\mathbb{N}})]_T| = 1$ . Then  $\pi(Y)$  is  $(<\mathbb{N})$ -scrambled, its closure is contained in the  $n$ -scrambled set  $\pi(2^{\mathbb{N}})$ , and Proposition 1.6 ensures that there is no perfect  $(n+1)$ -scrambled set, so the set of points in  $\pi(Y)$  that are also condensation points of  $\pi(Y)$  is as desired.  $\square$

Let  $\mathfrak{c}$  denote the cardinality of the continuum.

**Theorem 2.1** (ZFC +  $\neg \text{CH}$ ). *It is consistent that, for all  $n \geq 2$ , there is a homeomorphism  $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  with no  $(n+1)$ -scrambled perfect set but with a  $(<\mathbb{N})$ -scrambled set of cardinality  $\mathfrak{c}$  whose closure is  $n$ -scrambled.*

*Proof.* By [She99, Theorem 1.13], we can assume that there is an  $F_\sigma$  graph  $G$  on  $2^{\mathbb{N}}$  with a clique of cardinality  $\mathfrak{c}$  but no perfect clique. The proof of Theorem 1 therefore yields the desired result.  $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

**Theorem 2.2** (ZFC +  $\neg$ CH). *It is consistent that, for all  $n \geq 2$ , every Borel function on a Polish space with an  $n$ -scrambled set of cardinality  $\aleph_2$  has an  $n$ -scrambled perfect set.*

*Proof.* By [KS03, Proposition 3.4], we can assume that every Borel  $n$ -ary hypergraph on a Polish space with a clique of cardinality  $\aleph_2$  has a perfect clique. But  $n$ -ary Li–Yorke-ness is a Borel hypergraph.  $\square$

**Remark.** The proof of [KS03, Proposition 3.4] can be used to establish the analog of Theorem 2.2 for  $(<\aleph_2)$ -scrambled sets.

**Acknowledgements.** I would like to thank Alexander Kechris for encouraging me to utilize ChatGPT-5.6 Sol, which provided substantial assistance in the discovery and development of the results.

## REFERENCES

- [GGM21] Stefan Geschke, Jan Grebík, and Benjamin D. Miller, *Scrambled Cantor sets*, Proceedings of the American Mathematical Society **149** (2021), no. 10, 4461–4468.
- [Kec95] Alexander S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.
- [KS03] Wiesław Kubiś and Saharon Shelah, *Analytic colorings*, Annals of Pure and Applied Logic **121** (2003), no. 2–3, 145–161.
- [KV12] Wiesław Kubiś and Benjamin Vejnar, *Covering an uncountable square by countably many continuous functions*, Proceedings of the American Mathematical Society **140** (2012), no. 12, 4359–4368.
- [She99] Saharon Shelah, *Borel sets with large squares*, Fundamenta Mathematicae **159** (1999), no. 1, 1–50. MR 1669643

B. MILLER, 1008 BALSABWOOD DRIVE, DURHAM, NC 27705

Email address: glimneffros@gmail.com

URL: <https://glimneffros.github.io>